The Price of Exclusion: SME Set-Asides in Public Procurement

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Abstract

SME set-asides help protected firms by excluding their rivals. That exclusion has a price. I study a legal reinterpretation that expanded SME-only tendering in São Paulo's electronic procurement platform and use reverse-auction drop-out bids to recover type-specific cost distributions. The set-aside moves two margins in opposite directions: it removes non-SME bidders from the price-forming pool, while changing the protected SME pool through participation and composition. A structural decomposition shows that the protected-pool offset recovers only part of the lost competitive discipline. Removing non-SMEs raises simulated prices by 0.371/0.565 of the reference price in non-pharmaceutical/pharmaceutical procurement; the observed post-policy SME pool offsets -0.144/-0.256, leaving positive net effects. Static welfare losses equal 28.9/44.8 percent of the open-regime price at $\lambda = 0.30$. A 10 percent SME price preference keeps non-SMEs in the auction and has near-zero price effects in thick standardized markets, though it delivers less redistribution. The policy lesson is a frontier, not a ban on SME support: full exclusion is attractive only when the planner values additional SME surplus, access, or dynamic capacity gains enough to pay for the missing competitive discipline.

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1. Introduction

Procurement set-asides for small and medium-size enterprises (SMEs) are meant to widen access to public contracts. They do so by changing who is allowed to compete: some contracts are reserved for small firms, and larger firms are shut out. That design creates a basic tension. A set-aside may attract protected firms into the auction, yet it also removes the unprotected firms that would otherwise discipline price formation. The protected-pool offset pushes prices down; the loss of competitive discipline pushes them up. The welfare case depends on which force is larger.

I estimate this decomposition. The issue is not whether SME set-asides can raise procurement prices; a large empirical literature already shows that bidder-preference and set-aside programs can be costly (Marion, 2007; Krasnokutskaya, 2011; Athey et al., 2013; Nakabayashi, 2013; Hyytinen et al., 2018). The harder question is why they are costly, and what that says about policy design. A reduced-form price effect cannot separate exclusion from the change in SME participation and composition. The relevant comparison is therefore not SME support versus no support. It is support that preserves the price-forming pool versus support that removes rival bidders from it.

The empirical setting is São Paulo's centralized electronic procurement platform, the Bolsa Eletrônica de Compras (BEC). In late 2017 and early 2018, a legal reinterpretation shifted how the state applied Brazil's SME-only procurement rule. The old interpretation applied the eligibility threshold to the total value of a purchase notice; the new interpretation applied it item-by-item. That change mechanically exposed medical and hospital supplies (Group 65) to SME-only tendering because these purchases often combine many small items inside large purchase notices. The empirical cutoff is March 2018, when Group-65 public buyers began mass take-up of the new functionality. The trigger was legal and administrative rather than a medical-supply-specific policy initiative, and it changed admissibility rather than the production technology of the goods.

The first-stage facts fit the institutional account. After the cutoff, SME-only adoption in Group 65 rises sharply, non-SME participation falls, SME participation roughly doubles, and winning prices rise. A compact difference-in-differences benchmark against 76 never-treated product groups places the associated price movement at about 10–11 percent in the structural window. I use that reduced-form evidence as a sign, timing, and scale check, not as the main estimate. The central object is a structural decomposition, under explicit auction assumptions, of the price-forming forces that move when the policy removes non-SMEs and changes the active SME pool.

The auction format makes that decomposition feasible. BEC's main procurement format is the electronic *Pregão*, an English-reverse auction in which firms submit successively lower offers until only one remains. Under independent private values, losing bidders' drop-out prices reveal their costs (Haile and Tamer, 2003). This is the key identifying advantage of the setting, and also the main structural restriction. I use drop-outs to recover type-specific cost distributions for SMEs and non-SMEs, correct for auction-level scale heterogeneity in the spirit of Krasnokutskaya (2011), and simulate counterfactual auctions under observed equilibrium entry. The exercise is not a full entry equilibrium and not a reduced-form treatment-effect estimate: observed pre- and post-policy entry rates are taken as primitives, and the post-policy SME cost distribution is estimated from post-policy exits rather than derived from a formal selection model.

The decomposition uses three counterfactual prices. Let S_1 be the open auction with the pre-policy bidder pool, S_2 the SME-only auction holding the pre-policy SME pool fixed, and S_3 the SME-only auction with the observed post-policy SME pool. Then

$$p_{S_3} - p_{S_1} = \underbrace{p_{S_2} - p_{S_1}}_{\text{lost competitive discipline}} + \underbrace{p_{S_3} - p_{S_2}}_{\text{protected-pool offset}}.$$

The first term is the cost of removing non-SMEs from the price-forming pool. The second term is the offset created by the post-policy protected pool; it combines additional SME participation with changes in the composition of active SME bidders. This decomposition turns a set-aside price effect into a design question: how much of the cost comes from replacing competition by eligibility, and how much is recovered by the protected pool that the policy brings into the auction?

The main result is that the protected pool responds, but exclusion dominates. Removing non-SMEs while holding the pre-policy SME pool fixed raises simulated prices by 0.371 of the reference price in non-pharmaceutical supplies and 0.565 in pharmaceuticals. Replacing that fixed pool with the observed post-policy SME pool offsets part of the shock, by -0.144 and -0.256 respectively. The net effect remains positive: 0.227 in non-pharmaceuticals and 0.309 in pharmaceuticals. In absolute-magnitude terms, the exclusion component accounts for 72.0 and 68.8 percent of the two components. The interpretation is simple: the policy attracts and reshuffles SMEs, but it does not recreate the competitive discipline supplied by excluded non-SMEs.

This price-formation result changes the policy comparison. A full set-aside supports SMEs by removing their rivals. A price preference supports SMEs while leaving non-SMEs inside the auction. I use a 10 percent SME price preference as a static design benchmark: all firms remain eligible, entry and cost primitives are held fixed, and the scoring rule tilts winner selection toward SMEs. This is not a full implementation model of a legal reform. It is a comparison of two ways to support SMEs: preserving the price-forming pool versus deleting rival bidders from it. In thick, standardized non-pharmaceutical markets, the preference benchmark has near-zero price and welfare cost while the full set-aside generates a substantial welfare loss. The preference is not a free version of the set-aside; it delivers a smaller redistributive intervention. The correct comparison is a frontier. A social planner should prefer the set-aside only if the additional SME surplus is worth the allocative and fiscal cost of removing non-SMEs from the auction.

The strongest policy result is in thick standardized markets. In thinner and more heterogeneous pharmaceutical markets, the ranking depends on how the post-policy SME pool is modeled. The main specification estimates the post-policy SME cost distribution directly from post-policy exits; a strict invariance benchmark instead imposes that the SME cost distribution is the same before and after the policy. The non-pharmaceutical ranking is stable across these readings. The pharmaceutical ranking is not. I treat that difference as a scope condition rather than as a second headline: bidder exclusion is hardest to justify where the protected pool is thick enough for a preference rule to tilt allocation without removing competitive discipline, and most defensible, if at all, where the protected pool is thin, the good is heterogeneous, and the planner places high value on SME surplus.

The paper makes three contributions. First, it decomposes the price effect of SME set-asides into two opposing channels: the loss of competitive discipline from excluding non-SMEs and the protected-pool offset. Second, it uses the cost-revealing structure of electronic reverse auctions to estimate that decomposition from drop-out bids in a real procurement reform, linking a legal shock to structural price formation without relying on first-price bid-function inversion. Third, it recasts the set-aside-versus-preference comparison as a question of preserving competition inside the auction. This connects the empirical procurement-preference literature (Marion, 2007; Krasnokutskaya, 2011; Krasnokutskaya and Seim, 2011; Athey et al., 2013) to a public-finance welfare-weight interpretation in which transfers to SMEs, allocative waste, and the marginal cost of public funds are kept separate (Saez and Stantcheva, 2016).

The broader lesson is that eligibility is an expensive way to buy redistribution when excluded firms are the ones disciplining the auction. Set-asides can be defensible, but the defense must be explicit: the planner must value the additional SME surplus, market-access objective, or dynamic capacity gain enough to justify the allocative and fiscal cost of removing rival bidders. The natural benchmark is therefore not no SME support, but SME support that keeps competition in the room.

2. Setting, Data, and First-Stage Facts

The empirical episode is a legal change that shifted which firms could compete for eligible Group 65 items. Around the same cutoff, SME-only take-up rises, non-SME participation falls, SME participation rises, and prices move up. These facts establish timing and direction. They do not, by themselves, identify the structural price mechanism.

2.1. Legal shock

Brazil's federal SME statute, LC 123/2006, requires exclusive SME tenders for procurement items valued at R$ 80,000 or less. The 2014 amendment, LC 147/2014, made the SME-only rule mandatory rather than optional. For most goods on São Paulo's BEC platform, that rule already operated before the period studied here.

Group 65—medical, dental, and hospital equipment and supplies—was different for a legal reason. The prevailing interpretation applied the R$ 80,000 threshold to the total value of the purchase notice, not to each item or lot. Because medical-supply notices often bundle many small items inside large purchase orders, the total-notice interpretation kept many individually eligible items outside the SME-only default.

The relevant change was a generic legal reinterpretation. BEC enabled SME-only purchase-order functionality in COMUNICADO BEC nº 02/2017 on 18 July 2017 and mixed-exclusivity notices in COMUNICADO BEC nº 03/2017 on 20 November 2017. PGE-SP then issued Parecer Sub-G Cons. nº 151/2017 on 11 December 2017, holding that the R$ 80,000 threshold applies item-by-item regardless of the total value of the purchase offer. The empirical cutoff is March 2018, when Group 65 public buyer units began mass take-up of the functionality. TCE-SP aligned with the new interpretation on

16 May 2018, after the empirical cutoff, ratifying rather than initiating the operational sequence.

The institutional claim is narrow. The trigger was not a Group 65 market intervention and did not speak to medical-supply prices, supplier complaints, or technology. It was a general reading of procurement law that interacted with the way Group 65 notices were written. The cutoff shifts admissibility and the active bidder pool; it does not by itself identify a full entry model or rule out composition changes among SMEs. Online Appendix OA-A reports the institutional timeline and platform sample details.

2.2. Data and auction format

The data are administrative microdata from BEC, São Paulo's centralized electronic procurement platform. The raw extract contains 3.7 million bid-level observations across 82,569 items, 832,984 purchase orders, and 1,344 public buyer units. Group 65 accounts for roughly 27 percent of platform transactions in the window used here and is the treated group. The reduced-form benchmark compares Group 65 to 76 never-treated product groups.

The structural exercise uses Group 65 Pregão auctions in a symmetric 18-month window on each side of the March 2018 cutoff, from September 2016 through August 2019. The Pregão format is central: it is an electronic English-reverse auction in which the auction opens at a buyer reference price, firms submit lower offers, and firms exit as the clock falls. Under the independent-private-values interpretation used in the model, losing drop-out prices are cost observations.

For that reason, the structural sample restricts attention to Pregão auctions with at least 2 active firms and normalized bids $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$. The resulting sample contains 297,967 firm-auction observations across 97,993 auctions, split almost evenly between the pre-period (48,740 auctions) and post-period (49,253 auctions).

Each observation is classified by bidder type and product class. SME status is taken from the BEC administrative registry and validated against historical firm bidding records. I split Group 65 into non-pharmaceutical supplies and a pharmaceutical class covering CMED-regulated medicines and biologicals. This split is not a second treatment. It is a market-thickness distinction: the non-pharmaceutical segment is closer to a thick standardized market, while pharmaceutical procurement has fewer SME bidders and more within-class heterogeneity. The main policy claim is strongest in the former; the latter is used as a boundary case.

2.3. First-stage facts

Four facts matter. SME-only adoption rises in Group 65 after the cutoff; state-level adherence reaches 43 percent rather than full coverage. Non-SME participation falls. SME participation rises. Winning prices move up in the same period. Together, these facts are the reduced-form footprint of the price mechanism estimated below.

Table 1 reports the participation margins in the structural sample. In non-pharmaceutical Pregão auctions, the average number of SME bidders rises from 0.94 to 1.87, while non-SME participation falls from 2.68 to 1.50. In pharmaceutical auctions, SME participation rises from 0.55 to 1.22, while non-SME participation falls from 2.61 to 1.66. The post-period non-SME counts remain positive because take-up is incomplete, but both margins move in the direction required by the decomposition.

Prices move in the same period. Table 2 reports a compact difference-in-differences benchmark comparing Group 65 to never-treated product groups, with item and month fixed effects, auction-format and quantity controls, PBU controls, and item-clustered standard errors. The coefficient is written in a "DiD-in-reverse" convention: a negative coefficient on Group 65 interacted with the pre-period means that the open regime was cheaper, so the post-cutoff SME-only extension undoes that discount. Across windows, the log-price coefficient is between -0.108 and -0.148 with PBU controls; the 18-month benchmark is -0.113. Exponentiating the magnitude gives the 10–11 percent price bench-

Table 1: First-stage participation facts in Group 65 Pregão auctions

	Non-pharma	Pharma
Pre: SME bidders per auction	0.94	0.55
Pre: non-SME bidders per auction	2.68	2.61
Post: SME bidders per auction	1.87	1.22
Post: non-SME bidders per auction	1.50	1.66
Change in SME bidders	+0.93	+0.67
Change in non-SME bidders	-1.18	-0.95

Sample: Group 65 Pregão auctions in the 18-month window on each side of the March 2018 cutoff, after the structural filters $c_\epsilon \in (0, 3]$ and at least 2 active firms. Counts are average distinct bidders per auction by type and class. Post-period non-SME counts remain positive because the empirical SME-only take-up rate is 43 percent, not 100 percent.

mark used in the introduction.

Table 2: Reduced-form price benchmark

Window around cutoff	6 months	12 months	18 months
$g65 \times Pre$ on $\log p^{\text{final}}$	-0.148	-0.108	-0.113
Standard error	(0.014)	(0.013)	(0.012)
Observations	219,535	439,054	649,714

Estimates use item and month fixed effects, controls for sealed-bid format and log quantity, PBU controls, and item-clustered standard errors. The coefficient is reported in the same sign convention: a negative pre-period coefficient is the open-regime price discount that disappears after the SME-only extension. The DiD is used as a sign, timing, and scale benchmark; the structural sections estimate the mechanism.

The reduced form has a limited role. Group 65 differs from the pooled controls in several pre-period characteristics, and modern DiD estimators are less precise than the two-way fixed-effect benchmark. More importantly, the same legal shock both removes non-SMEs from the eligible pool and changes SME participation. A price regression can show that prices move at the cutoff; it cannot separate lost competitive discipline from the protected-pool offset. Online Appendix OA-B reports the balance, placebo, and alternative-DiD diagnostics behind this conservative reading. The rest of the paper uses the Pregão cost information to separate those margins.

3. Model and Identification

The empirical facts in Section 2 show participation and prices moving in the direction implied by the legal shock. To separate the two channels, I use a deliberately spare auction model. The model recovers type-specific costs from the auction format, treats observed entry as an equilibrium object, and simulates three counterfactual bidder pools. Its purpose is to decompose the price effect of exclusion, not to estimate a full entry equilibrium.

3.1. Auction environment

Consider a single procurement auction for one indivisible contract. Each firm i has type $k_i \in \{\text{SME}, \neg\}$ and draws a private supply cost c_i from a type-specific distribution F_c^k. The two distributions are allowed to differ. The buyer runs an electronic Pregão, an English-reverse auction in which the price clock moves downward from the buyer's reference price and firms remain active until they are no longer willing to supply at the current price. The last active firm wins and supplies the contract at the clearing price.

Under the independent-private-values interpretation, exit at cost is weakly dominant in this format (Haile and Tamer, 2003). A losing firm's drop-out price is therefore a direct observation of its willingness to supply. This is the key identifying advantage of the setting, but it is also the main structural restriction in the paper. In first-price procurement auctions, the researcher observes strategically shaded bids and must recover costs by inverting a bid function. Here the price-forming statistic can be recovered from exits only if drop-outs are informative about costs rather than about collusive rotation, budget constraints, or other non-cost stopping rules.

The winner is censored because the winner does not drop out; its cost is known only to be no higher than the second-lowest exit. The baseline estimator includes the winner's final price as a tight upper-bound observation, while Section 6 reports both a losers-only estimator and a Turnbull NPMLE that treats the winner as left-censored at that bound. The relevant test for the decomposition is not whether every observed exit is a literal engineering cost, but whether alternative treatments of censoring, common auction shocks, and bidder dependence overturn which bidder pool supplies the price-forming order statistic.

The expected transaction price in the clock interpretation is the expected second-lowest cost, $E[c_{(2)}]$, while the production cost of the allocation is the winner's cost, $E[c_{(1)}]$. This distinction matters later for welfare: $c_{(2)}$ determines the government's payment, while $c_{(1)}$ determines allocative efficiency.

3.2. Recovering cost primitives

The structural sample is organized by class, period, and bidder type. For each cell, loser drop-outs identify the corresponding empirical distribution of normalized costs. Costs are normalized by the buyer's reference price, so the object entering the simulation is a distribution of c_i/p^{ref}. Reference-price normalization absorbs the main cross-item scale differences, but it does not remove all auction-level heterogeneity. Two auctions in the same product class can differ in volume, delivery burden, regulatory documentation, or item-specific complexity in ways that move every bid in the auction together.

I therefore use the standard multiplicative auction-level heterogeneity correction of Krasnokutskaya (2011). Let

$$y_{it} = \log(b_{it}/p_t^{\text{ref}}) = a_t + e_{it},$$

where a_t is an auction-level scale component and e_{it} is the bidder-specific cost component. A method-of-moments variance decomposition estimates the between-auction and within-auction components, and a best-linear-predictor shrinkage removes $\hat{a}_t$ from each observation. The simulation uses the cleaned residual cost distribution $\exp(\hat{e}_{it})$. In the current estimates, the resulting Pregão intraclass correlations range from 0.36 to 0.59 in the main cells, with values comparable to procurement settings where this correction is standard.

Section 6 disciplines this cost-recovery step. It reports alternative treatments of winner censoring, a cross-modality check comparing Pregão drop-outs with a GPV recovery from Convite first-price bids (Guerre et al., 2000), dependence sensitivities, and bidder-pair coordination screens. These checks do not prove IPV. They ask the narrower question needed for this paper: whether the exclusion-dominant decomposition is mechanically produced by winner censoring, common auction-level scale heterogeneity, an obvious conflict between auction formats, or a post-policy increase in bidder clustering among the surviving SME pool. Online Appendix OA-C reports the corresponding quantile, KS, and censoring diagnostics.

3.3. Entry and the post-policy SME distribution

Entry is treated as observed equilibrium entry. For each class and period, I use the observed average numbers of SME and non-SME bidders per auction as the arrival rates for the simulation. In the open pre-policy regime, both SMEs and non-SMEs may enter. In the SME-only counterfactual, non-SMEs are removed from the eligible pool. The observed post-policy SME count is used for the post-policy SME-only scenario.

This choice keeps the identifying burden where the data can bear it. A full entry model would require assumptions on fixed costs, information, bidder-specific participation decisions, and how the legal shock changes expected profits. The legal shock directly identifies admissibility and observed participation, not a free-entry primitive. I therefore follow the reduced-form-entry spirit of Athey et al. (2011): observed pre- and post-policy bidder counts are equilibrium objects used in counterfactual price formation.

The same caution applies to the post-policy SME cost distribution. The baseline simulation estimates $F_c^{\mathrm{SME,Post}}$ directly from post-policy drop-outs and uses it as the distribution of active post-policy SME bidders. This object may reflect selection into the protected market, technology or sourcing changes within the SME segment, or other composition changes around the cutoff. The baseline does not derive that difference from a formal selection model. The strict-invariance benchmark instead imposes $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$. The comparison between these readings is a scope condition: the non-pharmaceutical policy ranking is stable, while the pharmaceutical ranking is not.

3.4. Counterfactual objects

The decomposition uses three simulated auction environments:

- S_1: open auction, pre-policy bidder pool. SMEs and non-SMEs enter at their observed pre-policy rates, with pre-policy cost distributions.
- S_2: SME-only auction, pre-policy SME pool. Non-SMEs are removed, while the SME arrival rate and SME cost distribution are held at their pre-policy values.
- S_3: SME-only auction, observed post-policy SME pool. Non-SMEs are removed, and SMEs enter with the observed post-policy arrival rate and the baseline post-policy SME cost distribution.

For each scenario, I draw bidder counts from the corresponding type-specific arrival process, draw costs from the recovered cost distributions, sort the draws, and record $c_{(1)}$ and $c_{(2)}$. The simulated price is $p_S = E_S[c_{(2)}]$, normalized by the reference price. The simulated production cost is $E_S[c_{(1)}]$.

The total set-aside price effect is

$$\Delta^{SA} = p_{S_3} - p_{S_1}.$$

It decomposes mechanically into two terms:

$$p_{S_3} - p_{S_1} = \underbrace{p_{S_2} - p_{S_1}}_{\Delta^{excl} \text{ lost competitive discipline}} + \underbrace{p_{S_3} - p_{S_2}}_{\Delta^{pool} \text{ protected-pool offset}}. \quad (1)$$

The first term holds the SME pool fixed and removes non-SMEs. It captures the price-forming cost of bidder exclusion: the second-order statistic is now drawn from the protected pool rather than from the full pool. The second term replaces the fixed pre-policy SME pool with the observed post-policy SME pool. It captures the extent to which additional protected participation and changes in the active SME composition offset the exclusion shock. This term can be negative, and in the data it is: the realized protected pool partially restores competition, but not enough to undo the price increase. I refer to it as the protected-pool offset because the legal shock does not separately identify a pure entry primitive from selection into the active protected-pool composition.

The labels in equation (1) are descriptive rather than behavioral. In an English-reverse IPV auction, surviving bidders need not shade bids differently when rivals are excluded; exit at cost remains weakly dominant. The lost-discipline component is therefore an order-statistic effect of admissibility, not evidence that SMEs bid less aggressively. This separates the channel studied here from first-price settings, where bidder preferences combine cost differences, order statistics, and strategic bid shading.

3.5. Scope of identification

The decomposition is mechanical once the cost distributions and bidder-count objects are specified. Its interpretation rests on four restrictions that the robustness section disciplines. First, drop-outs must be informative about costs or willingness to supply; the relevant failure is a systematic conduct or dependence change that makes the post-policy protected pool look expensive for reasons unrelated to the order statistic. Second, the post-policy SME distribution is an observed active-bidder distribution, not a primitive derived from a selection model. Third, average class-by-period arrival rates are a coarse summary of entry. Fourth, bidder-pair clustering means coordination must be treated as a first-order threat rather than a cosmetic robustness exercise.

These restrictions define the claim. The model identifies an auction-based decomposition of price formation under observed entry and explicit cost-recovery assumptions. It does not identify a full entry model, a complete model of SME selection, or a general conduct model for all procurement settings.

4. Exclusion, Entry, and Price Formation

The set-aside changes two margins at once. It removes non-SMEs from eligible auctions, while changing the protected SME pool through additional participation and composition. The protected pool responds, but the price effect is dominated by the loss of non-SME competitive discipline.

4.1. The protected pool responds

The participation facts rule out the simplest explanation for higher prices: protected firms do not fail to show up. In non-pharmaceutical Pregão auctions, the average number of SME bidders rises from 0.94 to 1.87 after the cutoff. In pharmaceutical auctions, it rises from 0.55 to 1.22. Non-SME participation moves in the opposite direction, falling from 2.68 to 1.50 in non-pharma and from 2.61 to 1.66 in pharma.

That response matters because it works against the price increase. If the SME pool were held fixed at its pre-policy size and composition, the set-aside would look more expensive than the realized post-policy environment. The central question is whether the realized protected pool restores enough price discipline to replace the excluded non-SMEs.

4.2. Exclusion dominates

Figure 1 puts the decomposition in one picture. The first bar, S_1, is the open pre-policy auction. The second, S_2, removes non-SMEs while holding the pre-policy SME pool fixed. The third, S_3, replaces that fixed pool with the observed post-policy SME pool. The jump from S_1 to S_2 is the lost competitive-discipline component. The decline from S_2 to S_3 is the protected-pool offset. The remaining gap between S_1 and S_3 is the realized price cost of the set-aside under the observed post-policy SME pool.

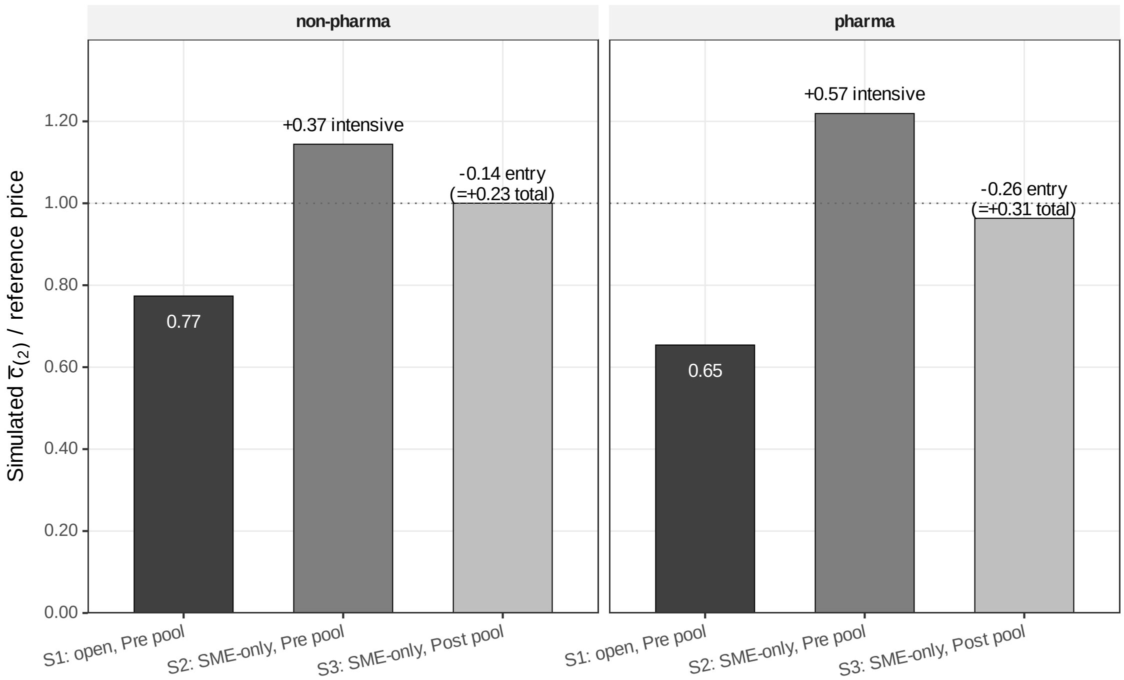

Figure 1: *Counterfactual price decomposition.* Bars report the simulated second-order statistic normalized by the buyer reference price. S_1 is the open pre-policy auction, S_2 removes non-SMEs while holding the pre-policy SME pool fixed, and S_3 allows the observed post-policy SME pool to replace the fixed pre-policy SME pool.

Table 3 reports the same decomposition numerically. In non-pharmaceutical markets, moving from S_1 to S_2 raises the simulated price from 0.774 to 1.144, an increase of 0.371 of the reference price. Replacing the fixed pre-policy SME pool with the observed post-policy SME pool reduces the simulated price by -0.144. The net effect remains positive, at 0.227. In pharmaceutical markets, the exclusion component is larger in levels: 0.565 of the reference price, partly offset by -0.256, leaving a net effect of 0.309.

The two normalizations answer different questions, but they point in the same direction. Relative to the net effect, the exclusion component is 164 percent in non-pharma and 183 percent in pharma because the protected-pool offset partially undoes the initial shock. Relative to the sum of absolute component magnitudes, exclusion accounts for

Table 3: Price decomposition: exclusion versus protected-pool offset

	Simulated price			Decomposition				
Class	S_1	S_2	S_3	$S_2 - S_1$	$S_3 - S_2$	$S_3 - S_1$	Excl./net	Abs. excl. share
Non-pharma	0.774	1.144	1.000	+0.371	-0.144	+0.227	164%	72.0%
Pharma	0.654	1.219	0.963	+0.565	-0.256	+0.309	183%	68.8%

S_1 is the open auction with the pre-policy bidder pool. S_2 is the SME-only auction holding the pre-policy SME pool fixed. S_3 is the SME-only auction using the observed post-policy SME pool. Prices are simulated $E[c_{(2)}]$ values normalized by the buyer reference price. "Excl./net" reports $(S_2 - S_1)/(S_3 - S_1)$. "Abs. excl. share" reports $|S_2 - S_1|/(|S_2 - S_1| + |S_3 - S_2|)$, where $S_3 - S_2$ is the protected-pool offset.

72.0 percent in non-pharma and 68.8 percent in pharma. The dominant force is the loss of the non-SME price-forming pool, not a failure of SMEs to respond.

4.3. The offset is not the main driver

The term $S_3 - S_2$ should not be read as a pure entry parameter. It combines additional SME participation with the empirical difference between the active SME bidders observed before and after the cutoff. That is why the paper treats $F_c^{\text{SME,Post}}$ as an observed post-policy active-bidder distribution rather than as the output of a formal selection model.

Two checks discipline this interpretation. First, under the strict-invariance benchmark that forces the post-policy SME distribution to equal the pre-policy one, the exclusion component remains the larger component: 85 percent in non-pharma and 79 percent in pharma. Second, the Turnbull cost estimator gives the same qualitative message, with exclusion shares of 74.0 percent in non-pharma and 82.0 percent in pharma on the absolute-share normalization. Online Appendix OA-C and OA-D report the underlying censoring, primitive stability, and mechanism tables.

Composition matters most for the policy ranking in pharma. In pharmaceuticals, 61.9 percent of post-policy SME firms are new relative to the pre-period pool and they account for 37.8 percent of post-policy SME bids. In non-pharma, the corresponding bid share is 23.0 percent. The composition shift is therefore largest precisely in the class where the welfare ranking is model-sensitive. The robust price-formation result is that bidder exclusion is the larger force, especially in the thicker non-pharmaceutical market.

4.4. Interpretation

The mechanism is not that protected firms strategically bid less aggressively after the set-aside. In the maintained Pregão/IPV model, bidders exit at cost. The mechanism is that the set-aside changes which cost order statistic forms the price. In the open auction, non-SMEs remain in the pool and discipline the second-lowest cost. In the SME-only auction, that discipline is removed. The post-policy protected pool pushes back against the loss, but it does not recreate the full-pool benchmark.

This price-formation result carries into welfare. A policy that supports SMEs by excluding non-SMEs buys redistribution by weakening the auction's price-forming pool. A policy that supports SMEs while leaving non-SMEs in the auction works through a different channel: it can tilt allocation toward SMEs while preserving the bidders that discipline the price.

5. Welfare and Policy Design

The price decomposition maps naturally into welfare. The simulated second-order statistic, $c_{(2)}$, determines the government's payment; the winner's cost, $c_{(1)}$, determines allocative efficiency. A set-aside can raise payments without turning the full price increase into deadweight loss: part of the extra payment is a transfer to the protected winner. The social cost is the real production inefficiency from assigning the contract to a higher-cost firm plus the tax distortion from financing the additional public outlay.

5.1. From price formation to welfare

Let p^V denote the expected procurement price under policy V, and let $c^V_{(1)}$ denote the winner's production cost. Relative to the open benchmark S_1, the static per-auction welfare loss from the SME-only set-aside V_0 is

$$\text{Loss}(V_0) = \underbrace{c_{(1)}^{V_0} - c_{(1)}^{S_1}}_{\text{DWL}_{\text{alloc}}} + \underbrace{\lambda\left(p^{V_0} - p^{S_1}\right)}_{\text{MCPF distortion}}. \tag{2}$$

The baseline uses $\lambda = 0.30$. The remaining component of the extra payment is an implicit transfer to SME bidders; it becomes a social benefit only if the planner places extra welfare weight on SME surplus.

5.2. Set-aside versus preference

Table 4 compares the full set-aside with a non-exclusionary benchmark. The benchmark, denoted V_3, is a static open-auction scoring rule: SMEs receive a 10 percent price preference for winner selection, all firms remain eligible, and the government pays the actual winning bid. I hold recovered cost distributions and entry primitives fixed. This counterfactual isolates the value of preserving the non-SME price-forming pool; it is not a full model of how firms or procuring units would adapt to a new legal regime.

Table 4: Welfare cost of exclusion and the price-preference benchmark

	Full SME set-aside V_0				10% preference V_3	
Class	Δ_{gov}	$\text{DWL}_{\text{alloc}}$	MCPF	Loss / p_{S_1}	Δp	SME win gain
Non-pharma	0.247	0.148	0.074	28.9%	-0.004	4.3 pp
Pharma	0.298	0.207	0.089	44.8%	+0.002	1.4 pp

V_0 is the full SME-only set-aside under the observed post-policy SME pool. Δ_{gov}, $\text{DWL}_{\text{alloc}}$, MCPF, and Δp are reported in reference-price units. MCPF uses $\lambda = 0.30$. V_3 is the static 10 percent preference benchmark described in the text. SME win gain is relative to the open benchmark S_1.

The full set-aside is costly on both welfare margins. In non-pharmaceuticals, it raises government payments by 0.247 of the reference price, of which 0.148 is allocative waste and 0.074 is the MCPF distortion. The total welfare loss is 28.9 percent of the open-regime price. In pharmaceuticals, the corresponding welfare loss is 44.8 percent. The lower endpoints of the bootstrap intervals remain economically large (20.5 percent in non-pharmaceuticals and 34.9 percent in pharmaceuticals).

The preference benchmark works differently. Non-SMEs continue to participate and remain available to discipline the price-forming order statistic. Holding the active bidder pool and recovered costs fixed, the simulated price effect is essentially zero: -0.004 of the reference price in non-pharmaceuticals and $+0.002$ in pharmaceuticals. The preference is not a free version of the set-aside; it delivers less redistribution. The SME win-rate rises by 4.3/1.4 percentage points rather than being forced to 100 percent. The comparison is therefore a frontier: set-asides buy more redistribution by removing rival bidders, while preferences buy less redistribution while preserving the price-forming pool.

The low-cost region is not a knife-edge at 10 percent. In the preference grid, welfare costs remain inside Monte Carlo noise up to 15 percent in non-pharmaceuticals and 25 percent in pharmaceuticals, and even a 30 percent preference costs only 4.99/1.92 percent of p_{S_1}. These are static comparisons, not implementation forecasts, but they show why exclusion is an extreme point on the policy frontier in thick standardized procurement. Online Appendix OA-E reports the full preference grid and the welfare-ranking sensitivity across the λ grid.

5.3. How much must the planner value SME surplus?

The distributional case for exclusion can be stated as a welfare-weight condition. Following the social marginal welfare weight logic in Saez and Stantcheva (2016), let w^{SME} be the planner's weight on one real of SME producer surplus relative to one real of general surplus. The planner is indifferent between the set-aside and the 10 percent preference when

$$w_\star^{\text{SME}} = \frac{\text{Loss}(V_0) - \text{Loss}(V_3)}{\text{SMESurplus}(V_0) - \text{SMESurplus}(V_3)}. \tag{3}$$

Because V_3 has near-zero welfare cost in the static benchmark, the numerator is essentially the deadweight cost of exclusion. The denominator is the additional SME surplus created by moving from the preference benchmark to the full set-aside. Under the main specification, the implied indifference weights are 2.42 in non-pharmaceuticals and 2.61 in pharmaceuticals. In words, the planner must value an extra real of SME surplus at more than twice a real of general surplus to prefer excluding rival bidders over preserving them with this preference rule.

Dynamic benefits of SME market access, learning, or capacity building would enter this comparison as additional benefits of the set-aside; they are not identified by the static auction exercise. That limitation is also where the market-structure scope condition matters. In non-pharmaceuticals, the preference rule continues to dominate the set-aside under both the main and strict-invariance readings. In pharmaceuticals, the ranking is sensitive to how the post-policy SME pool is interpreted; under strict invariance the implied weight falls to 0.7. The argument against exclusion is strongest where the protected pool is thick enough that redistribution can be achieved without changing the price-forming bidders.

5.4. Scope conditions

These estimates do not imply that set-asides are never defensible. They imply that the defense must be explicit. Exclusion becomes more plausible when the protected pool is thin, when the planner wants protected market access rather than merely more marginal SME awards, when dynamic SME capacity gains are expected to be large, or when the planner places a high social weight on SME surplus. Those conditions are not identified by the legal threshold alone. They are market-structure conditions.

Non-pharmaceutical Group-65 procurement is the hard case for the set-aside: the SME pool is thick, goods are standardized, and the preference benchmark preserves competition while delivering SME wins at almost no static welfare cost. Pharmaceuticals are the boundary case: the SME pool is thinner, composition changes more under the policy, and the welfare ranking depends on how the post-policy SME distribution is interpreted. The design principle is therefore conditional rather than universal. Exclusion should be used only when the planner is prepared to pay for the missing competitors; otherwise, SME support should keep those competitors in the auction.

6. Robustness to Core Threats

The robustness checks follow the empirical chain. The legal shock must line up with the timing of prices; drop-out bids must recover the cost objects needed for the order-statistic comparison; the post-policy SME pool must not be the sole source of the result; coordination must not intensify after restriction; and the preference benchmark must not be a mechanical artifact. The question is whether any of these threats overturns the conclusion that bidder exclusion is the main source of the price increase.

6.1. Reduced-form timing and scale

The reduced-form evidence is used for timing and scale, not for the structural decomposition. The 18-month benchmark in Table 2 is -0.113 in the paper's DiD-in-reverse convention, implying a 10–11 percent price movement. The estimate is stable across the 6-, 12-, and 18-month windows. Excluding the BEC enablement months before the mass-take-up cutoff changes the coefficient from -0.109 to -0.087 with a standard error of 0.012, and placebo cutoffs before the reform are smaller in magnitude (-0.013, -0.030, and -0.034) than the real-cutoff estimates.

The conservative reading remains necessary. Group 65 is not perfectly balanced against the pooled controls: 7 of 9 pre-period covariates exceed the 0.10 normalized-difference threshold, and 4 exceed 0.25. Modern DiD variants also reduce precision: the BJS imputation estimate is -0.0560 and the Callaway–Sant'Anna estimate is -0.0165. The reduced form therefore validates the episode's sign and timing; it does not identify the exclusion and protected-pool channels. Online Appendix OA-B reports the balance table, placebos, and alternative estimators.

6.2. *Cost recovery and auction primitives*

The cost-recovery concern is that the decomposition could be an artifact of winner censoring, common auction-level shocks, or bidder dependence. The diagnostics do not support that reading. Alternative treatments of winner censoring give similar net price effects: 0.275/0.259/0.246 in non-pharmaceuticals under losers-only/all-bidders/Turnbull inputs, and 0.347/0.308/0.357 in pharmaceuticals. The Turnbull within-auction share remains large (74.0 percent in non-pharmaceuticals and 82.0 percent in pharmaceuticals), so the result is not driven by treating the winner's final bid as an exact cost observation.

The other diagnostics point the same way. The auction-level heterogeneity correction removes common scale shocks before simulation. A Convite first-price GPV recovery and the Pregão drop-out recovery line up in the key pharmaceutical non-SME pre-period cell after that correction. A Gaussian-copula relaxation allows within-auction cost correlation up to $\rho_c = 0.3$; across that grid, the within-auction share moves by less than 5 percentage points and the total effect by less than 10 percent. Tightening or loosening the $c_\epsilon \leq 3$ filter and using 18-, 12-, or 6-month windows also leave the within-auction component as the larger component in both classes. Online Appendix OA-C reports the cost-distribution quantiles and primitive-stability diagnostics behind these summaries.

6.3. *Protected-pool composition*

The main modeling threat is that $S_3 - S_2$ mixes additional SME bidders with a changed post-policy SME cost distribution. The strict-invariance benchmark sets $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$ while keeping observed post-policy SME entry counts. The total price effect remains positive: 0.29 in non-pharmaceuticals and 0.47 in pharmaceuticals. The within-auction shares rise to 85 and 79 percent. Composition therefore matters, but it is not what makes the exclusion component dominate the price decomposition.

Composition does matter for the policy ranking in pharmaceuticals. Post-policy pharmaceutical SME participation is heavily composed of new firms: 61.9 percent of post-period pharma SME firms are absent from the pre-period pool, and they account for 37.8 percent of post-period pharma SME bids. In non-pharmaceuticals, new firms account for 23.0 percent of post-period SME bids. The pharmaceutical welfare ranking is therefore conditional, while the non-pharmaceutical ranking is the more stable policy result.

6.4. Coordination

Coordination is the conduct threat that would most directly contaminate the drop-out interpretation. The data show bidder clustering: Conley-style close-pair screens (Conley and Decarolis, 2016) reject the independent-draw null in all four class-by-period cells, and persistent-pair diagnostics in the spirit of Bajari and Ye (2003) flag excess repeated co-bidding. The relevant question is whether clustering increases after non-SMEs are removed. It does not. The Conley realized close-pair share is essentially flat in non-pharmaceuticals (16.9 to 16.8 percent) and falls in pharmaceuticals (27.6 to 24.4 percent). The Bajari–Ye T_1 observed-to-null ratio also falls in both classes. Residual baseline clustering remains a limitation, but the strongest alternative story, a post-restriction coordination shock among surviving SMEs, is not supported by the diagnostics. Online Appendix OA-D reports the coordination-screen table.

6.5. Policy benchmark

The price-preference comparison is a static design benchmark, so the relevant robustness checks are mainly mechanical. The non-pharmaceutical ranking remains $V_3 \succ V_0$ over $\lambda \in [0.15, 0.45]$ under both the main and strict-invariance specifications; the pharmaceutical ranking remains conditional on the treatment of the post-policy SME pool. The result is also not an artifact of exact Vickrey equivalence. A Maskin–Riley first-price upper-bound adjustment (Maskin and Riley, 2000) is at most 5 percent of the reference price, while the simulated V_0–V_3 price gap is 22–26 percent. Online Appendix OA-E reports the preference grid and welfare-weight sensitivities.

The checks leave a narrow claim. The result does not depend on the reduced-form DiD carrying the whole design, on a single treatment of winner censoring, or on holding the post-policy SME pool fixed. When the excluded bidders are the ones disciplining the order statistic, set-asides are costly because they replace competition with eligibility. The protected-pool response offsets part of that cost, but it does not replace the competitive discipline that the policy removes.

7. Conclusion

SME set-asides are usually evaluated as a question of access: do protected firms receive more public contracts? I evaluate the instrument as a question of market design. A set-aside supports SMEs by changing eligibility, and changing eligibility changes price formation. It removes some bidders from the auction at the same time that it may attract protected firms into the auction. The welfare case depends on the balance of those forces.

The evidence from São Paulo's procurement reform points to that balance. The protected SME pool responds after the set-aside expands, but the response does not replace the competitive discipline supplied by excluded non-SMEs. The main price effect comes from the restricted auction's order statistic: once non-SMEs are removed, the price-forming bidder is drawn from a thinner and different pool. Participation and composition partly offset the shock, but they do not undo it. The mechanism is therefore not that SMEs simply bid less aggressively. It is that an exclusion rule changes who remains available to discipline the price.

This changes the relevant policy comparison. The benchmark is not SME support versus no SME support. It is exclusionary support versus support that preserves competition inside the auction. A full set-aside is a high-cost, high-redistribution instrument because it forces awards toward SMEs by removing rival bidders. A price preference is lower redistribution, but it keeps non-SMEs in the auction and allows them to remain price-forming. In the static benchmark for thick standardized markets, preserving competition has much lower welfare cost than removing it.

The conclusion is conditional. In thin or heterogeneous markets, especially where the post-policy protected pool changes substantially, the case for exclusion can be stronger and the ranking can become model-sensitive. That is the boundary of the design principle. A planner who chooses bidder exclusion should be explicit that the additional SME surplus, market-access objective, or dynamic capacity gain is worth the allocative and fiscal cost of taking rival bidders out of the auction.

The broader lesson is simple: eligibility is an expensive substitute for competition when the excluded firms are the ones disciplining the auction. Procurement policy should therefore treat full set-asides as the instrument that requires justification, not as the default form of SME support. The natural benchmark is support that keeps rival bidders in the auction.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The structural sample is derived from BEC administrative microdata accessed through a research agreement with the Secretaria da Fazenda e Planejamento do Estado de São Paulo. The raw administrative records are not publicly redistributable under the agreement. Replication materials with code, generated tables, figures, and non-confidential derived outputs will be made available in a public repository, subject to the data-use restrictions.

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